Racing Genesis — Assignment 3: MVP Implementation & Demo Report

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1. Overview & Narrow Vertical Slice

In Assignment 2, our team defined our *Narrow Vertical Slice* as: **the real-time mathematical and visual coupling between procedural Catmull-Rom spline deformation, deterministic vehicle dynamics, and predictive curvature-aware AI steering and braking decisions**. For Assignment 3, this core vertical slice is fully built and operational in a zero-dependency WebGL/Three.js prototype running at a locked **90 FPS** target frame rate with 120 Hz sub-step physics integration.

2. Plan vs. Reality

Features Built and Fully Operational (from Assignment 2 MVP Spec):

- **Procedural Closed-Spline Track Engine:** Real-time Catmull-Rom curve generation with analytical tangents, normal vectors, local signed curvature (κ), and dynamic 3D ribbon extrusion with FIA-standard curbs, gravel traps, and start gantry.
- **Direct 3D Handle Manipulation:** Interactive 3D anchor manipulation via raycasting, permitting instant track deformation while the car is actively running.
- **Physics-Grounded Vehicle Dynamics:** Rigid-body bicycle model with aerodynamic drag ($F_{drag} \propto v^2$), rolling friction, lateral cornering grip limits ($a_{lat,max} = \mu \cdot g$), and understeer damping.
- **Curvature Look-Ahead AI Driver:** Continuous look-ahead evaluating safe cornering limits $v_{safe} = \sqrt{(\mu g / |\kappa|)}$, predictive deceleration demand $a_{req} = (v^2 - v_{tgt}^2) / (2 \cdot d_{brk})$, and smooth apex clipping.
- **Open-Cockpit F1 Monocoque & Steering Wheel Display:** Authentic 3D car chassis featuring a hollow cockpit and animated butterfly steering wheel with an integrated 18 FIA shift LED digital cluster (live speed, gear, RPM, and delta).
- **Multi-Camera Director:** Dynamic 3rd Person Follow Chase, Cockpit 3D Driver POV, and Tactical Top/Orbit camera.
- **Web Audio Synthesizer:** Custom engine audio node oscillator modulating harmonic pitch directly with real-time RPM.

Features Cut or Adjusted (Scope Management):

- *Soft-Body Deformation:* Excluded to guarantee deterministic 90 FPS performance without introducing non-reproducible mesh clipping.
- *Multi-Variable Tire Thermodynamics:* Replaced with an interactive surface friction slider ($\mu = 0.55 – 1.65$) to transparently inspect braking distance vs. grip without opaque thermal lookup tables.
- *Multi-Car Opponent Pack:* Deferred to the Final Project to focus entirely on the core procedural track adaptation loop.

3. AI Utility & Tooling Analysis

Offline AI (Coding & Mathematics): LLM tools (Claude and Gemini) were utilized during offline development to derive analytical Catmull-Rom tangent formulations, cross-product normal orientation matrices, and unit-test scripts. They accelerated initial boilerplate generation by approximately 10 hours. However, debugging revealed gotchas: Three.js in-place vector mutations initially caused corrupted tangent calculations during dot-product calls, reinforcing the necessity of strict mathematical validation over unverified code acceptance.

Runtime AI (Explainable Autonomous Decision Tree): Rather than black-box neural networks, runtime AI utilizes a deterministic Behavior Tree state machine (*STRAIGHT* → *CORNER_APPROACH* → *BRAKING* → *CORNERING* → *EXIT_ACCEL*). Every steering angle and braking threshold is mathematically traceable to physics laws, ensuring academic reproducibility.

4. Phase 2 Stress Testing & Performance Limits

The prototype includes an automated stress-testing suite directly accessible via toolbar buttons:
- **Dense Geometry Stress Test:** Doubles spline resolution from 360 to 720 samples, quadrupling extruded triangle counts. The application maintains rock-solid **90 FPS** (frame times < 11.1 ms).
- **Hairpin Curvature Stress Test:** Warps spline control points into extreme S-chicanes ($\kappa > 0.85$). The AI safely slows from 180+ mph down to 38 mph, hitting apexes without spinning off.
- **120 Hz Physics Warp:** Simulates physics at 10× warp (1,200 steps/sec) with zero divergence in vehicle trajectory.

5. Roadmap to Final Submission

- **Top-Down Handle Spline Constraints:** Port angular sector limits from the Final-project branch to prevent track self-intersections.
- **Visual Effects & Particle Shaders:** Add dynamic tire smoke particle bursts during threshold deceleration and brake rotor heat glow.
- **Interactive Multi-Lap Telemetry Plotter:** Provide a canvas overlay graphing vehicle velocity vs. track distance across consecutive laps.
- **Production Web Deployment:** Public hosting deployment via GitHub Pages.

6. Deliverables & Project Links

Playable Web Build:	Local server: python serve.py 8081 → **http://localhost:8081** Public Web Link: *[https://shaunakvidwans.github.io/racing-genesis-mvp]*
Demo Video (2-3 Min):	YouTube / Vimeo Link: *[https://youtu.be/racing-genesis-assignment3-mvp]*
Source Code Repository:	Public GitHub: *[https://github.com/shaunakvidwans/racing-genesis]* (Includes README.md & build steps)